# Hierarchical Domain Adaptation Framework for Disparity Estimation in Optical Satellite Stereo Imagery: Bridging Spatiotemporal-Sensor Heterogeneity

Guangbin Zhang, Yonghua Jiang, Shaodong Wei, Yunming Wang, Jie Chu, Meilin Tan, and Zhiwei Li

***Abstract*— Deep learning-based disparity estimation methods have demonstrated significant potential in optical satellite stereo image applications. However, learning-based methods remain susceptible to domain shifts caused by spatiotemporal variations and stereo-sensor heterogeneity. To address these challenges, we propose a hierarchical domain adaptation disparity estimation (HDADE) framework for optical satellite stereo images. HDADE was structured with a four-stage technique pipeline to improve the training data quality and diversity, explicitly align the spectral and stereo distribution, implicitly enhance the robustness of feature extraction and matching, and directly facilitate feature alignment with the target domain. This hierarchical framework systematically mitigates disparity estimation accuracy degradation in cross-domain scenarios. Cross-spatiotemporal and cross-payload generalization experiments were conducted based on the WHU_Stereo and US3D datasets. The experimental results show that HDADE significantly outperformed other advanced methods and possessed plug-and-play versatility. Notably, greater domain shift scene transfer experiments indicated that, with limited annotation data, HDADE has the potential for large-scale automatic applications.**

***Index Terms*— Deep learning, disparity estimation, domain adaptation, hierarchical framework, satellite stereo images.**

## I. Introduction

In THE field of photogrammetry and remote sensing, obtaining dense and accurate disparity from stereo images (dense stereo matching/disparity estimation) has been a long-standing and classic research topic [1], [2], [3], [4]. Optical satellite stereo image disparity estimation (OSSIDE) can significantly contribute to the rapid large-scale 3-D reconstruction of ground scenes from satellite images. Therefore, this technology holds substantial application potential in areas such as 3-D smart cities, virtual reality, and digital Earth modeling [5].

Traditionally, dense stereo matching is conducted in several steps: matching cost computation, cost aggregation, disparity computation and optimization, and disparity refinement [6], [7], [8]. In the field of computer vision, meticulously annotated dense stereo matching datasets developed in recent years have propelled learning-based methods to achieve significant progress [9], [10], [11]. These advancements have facilitated the development of end-to-end dense stereo matching techniques for satellite remote sensing images [12], [13], [14], [15]. Owing to superior feature extraction and matching capabilities, deep learning-based disparity estimation methods have become the mainstream approach across various disparity estimation benchmarks in both computer vision and remote sensing fields.

Recent research on learning-based disparity estimation initially focused on incorporating deep learning into the various stages of classical disparity estimation methods [16], [17]. With the advancement in computer hardware, software, and the expansion of annotated data, end-to-end methods have emerged a new paradigm in the field of disparity estimation. DispNet [10] utilizes an encoder–decoder 2-D convolutional architecture for two-view feature extraction and similarity computation. PSMNet [12] uses spatial pyramid pooling and a stacked 3-D encoder–decoder architecture to extract contextual information and aggregate costs. STTR [18] integrates self-attention and cross-attention mechanisms to model the correlation between features from the two views. These end-to-end methods have substantially enhanced the accuracy of disparity estimation. Furthermore, in the fields of photogrammetry and remote sensing, BGA-Net [19] leverages multitask modules, HMSM-Net [20] employs a pyramid multiscale architecture, and $S^2$Net [21] combines data augmentation techniques to achieve superior dense matching performance on various stereo satellite image datasets.

Significant progress has been achieved in the development of learning-based disparity estimation algorithms. However,

Received 18 March 2025; revised 6 May 2025; accepted 22 May 2025. Date of publication 29 May 2025; date of current version 16 June 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42301424 and Grant 42171341. *(Corresponding author: Yonghua Jiang.)*

Guangbin Zhang, Yonghua Jiang, and Yunming Wang are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: zhanggb1997@whu.edu.cn; jiangyh@whu.edu.cn; 595589080@qq.com).

Shaodong Wei is with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: weishaodong@whu.edu.cn).

Jie Chu is with Xiangyang Institute of Surveying and Mapping, Xiangyang 441003, China (e-mail: JieChu.whu@qq.com).

Meilin Tan is with Inner Mongolia Autonomous Region Surveying and Mapping Geographic Information Center, Hohhot 010050, China (e-mail: tanmeilin@whu.edu.cn).

Zhiwei Li is with the 91001 Unit, Chinese People's Liberation Army, Ürümqi 100841, China (e-mail: lzw_cbs@163.com).

Digital Object Identifier 10.1109/TGRS.2025.3574984

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

the performance of these algorithms largely depends on two key factors: 1) the availability of a large amount of high-quality satellite stereo image pairs with disparity ground truth [22] and 2) the consistency of data distribution between the testing and training datasets [23], [24]. In practical applications of satellite imagery disparity estimation, it is challenging to construct a sufficiently large dataset of satellite stereo images with high-quality disparity [25], [26]. With respect to the latter factor, maintaining data distribution consistency between stereo images and the training data is difficult due to inconsistencies in satellite imaging payloads and spatiotemporal data [1], [23]. This domain shifting phenomenon in OSSIDE tasks can primarily be attributed to the following three aspects: 1) scale differences due to varying satellite payloads with different ground sampling distances; 2) radiometric differences due to variations in satellite imaging spatiotemporal and atmospheric conditions; and 3) stereo pair time lags and disparity distribution differences caused by varying stereo imaging modes. As illustrated in Fig. 1, the performance of existing learning-based disparity estimation networks significantly degrades in practical applications due to the constraints imposed by the two key factors mentioned above.

Inspired by the two aforementioned key constraints, this study introduces a hierarchical domain adaptation disparity estimation (HDADE) framework for optical satellite stereo images to enhance the usability of cross-domain OSSIDE applications. HDADE aims to construct a systematical end-to-end domain adaptation pipeline through four levels: data-level (DL) data optimization and coordinated augmentation, image-level joint spectral-disparity fitting, feature-level (FL) domain invariant representation adversarial learning, and instance-level dynamic self-supervision learning based on pseudo-label. Through data- and image-level technical components, the HDADE ensures training data quality while performing explicit data augmentation and spectral-disparity distribution alignment. Furthermore, feature- and instance-level technical components significantly enhance the domain sensitivity of network and achieve direct feature distribution alignment.

The main contributions of this study are as follows.

1) We propose an HDADE framework for end-to-end methods called HDADE. To the best of our knowledge, our study is the first to explore the generalization of OSSIDE network for cross-domain data.
2) HDADE addresses performance degradation caused by domain shifts through: DL data optimization and augmentation (DL-DOA), image-level spectral-disparity fitting, FL spectral feature and stereo cost adversarial (FL-SSAdv), instance-level pseudo disparity construction, and dynamic self-training. The framework systematically mitigates the performance degradation on cross-domain data.
3) Experiments were conducted using various disparity estimation networks, with cross-payload and cross-spatiotemporal datasets. The experimental results demonstrate that HDADE can effectively enhance their generalization with cross-domain data and can be plug-and-play integrated into end-to-end disparity estimation methods.

The remainder of this article is organized as follows. Section II introduces end-to-end disparity estimation methods and relevant domain adaptation studies. Section III presents the proposed HDADE domain adaptation framework. Section IV describes the experimental data and setup employed in this study. Section V evaluates the performance of the cross-domain disparity estimation methods. Section VI provides a related discussion. Section VII concludes the findings of this article.

## II. Related Works

### A. End-to-End Disparity Estimation Method

Existing end-to-end disparity estimation methods can be broadly classified into dense matching networks based on 2-D CNN, 3-D CNN, and transformer architectures according to their feature encoding and cost construction methods. Dense matching networks based on 2-D CNN architectures primarily use an encoder–decoder architecture to expand the receptive field and fully utilize contextual information in the stereo images. Additionally, these networks introduce a feature correlation layer to encode the similarity of the left and right features, thereby achieving pixel-level end-to-end disparity regression. Following this paradigm, Mayer et al. [10] designed a milestone end-to-end disparity estimation network called DispNet-C. Many scholars have optimized disparity estimation networks based on this basic paradigm by incorporating mechanisms such as residual refinement [27], [28], [29] and multitasking strategies [13], [30], [31], [32].

Disparity estimation networks based on 3-D CNN architectures explicitly construct geometric encoding within the network. Specifically, these networks use a 4-D cost volume combined with 3-D convolution for cost aggregation to describe the correlation between the left and right views. A differentiable soft-argmin operator is ultimately used for the disparity regression. Based on this concept, Kendall et al. [33] proposed the pioneering GC-Net. Subsequent researchers have continuously enhanced the advantages of disparity estimation networks based on 3-D CNN architectures by incorporating spatial pyramids and hourglass modules [12], cascade cost construction strategies [34], unimodal cost volume constraints [35], and semi-global guided aggregation modules [14].

Disparity estimation networks based on transformer architectures mainly use self-attention and cross-attention mechanisms to model the correlation between the left and right views. STTR [18] combines an end-to-end approach with transformer structures to perform stereo matching. Based on the STTR paradigm, Guo et al. [36] further add a contextual enhancement path to strengthen the acquisition of long-range global information. Lou et al. [15] combine 3-D convolution architectures to conduct intraevidence fusion based on multilevel predictions and interevidence fusion using various stereo matching methods.

In the fields of photogrammetry and remote sensing, BidirEPNet [37] improves on PSM-Net to ensure its suitability for satellite stereo image pairs. BGA-Net [19] simultaneously performs disparity estimation and semantic segmentation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

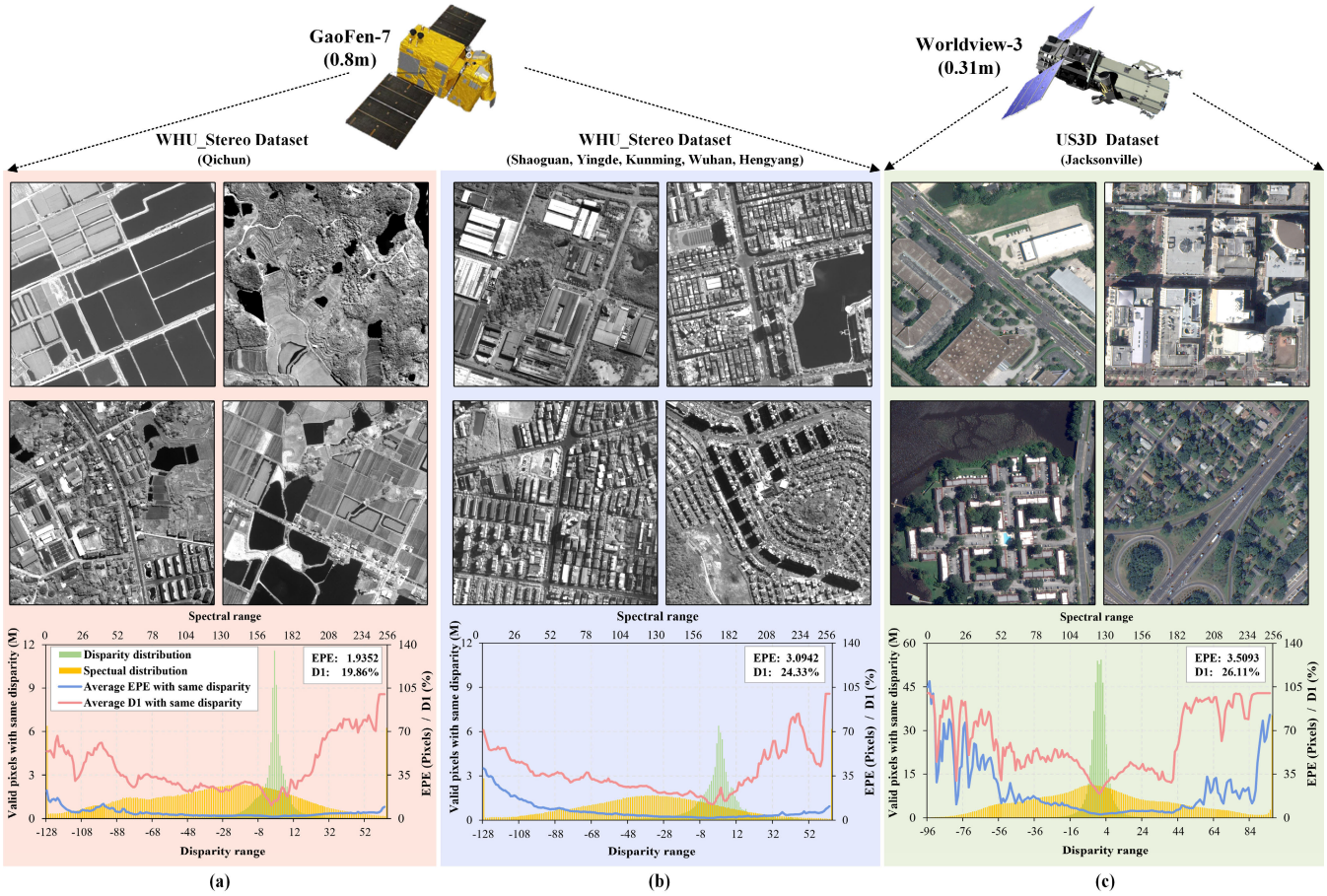



Fig. 1. Differences in satellite images and the decline in disparity estimation accuracy under different geographical locations and sensor imaging conditions. Images, spectral brightness distribution, and stereo disparity distribution from (a) Qichun in the WHU_Stereo dataset; (b) Shaoguan, Yingde, Kunming, Wuhan, and Hengyang in the WHU_Stereo dataset; and (c) Jacksonville in the US3D dataset. Additionally, HMSM-Net [20] was trained on the data (a) and evaluated the disparity estimation accuracy on test sets from (a) to (c). The accuracy evaluation scores and stereo disparity-based accuracy distribution are presented in the line charts.

tasks, thereby enhancing feature sharing and complementary optimization of results through a bidirectional attention mechanism. $S^2$Net [21] adopts a nonparallel multitask execution framework to generate more accuracy increments between disparity estimation and semantic segmentation tasks, and it also incorporates a data augmentation mechanism to improve the data diversity and model robustness. HMSM-Net [20] constructs a pyramid-shaped cost volume to robustly match contextual features at different receptive field levels and employs a hierarchical cost fusion method for adaptive cross-scale cost volume fusion and output.

The aforementioned disparity estimation methods have achieved significant performance improvements across various disparity estimation benchmarks and greatly promoted the application of end-to-end disparity estimation algorithms. However, these advanced disparity estimation methods do not generalize well to data with domain shifts [38]. In particular, domain shifts are pronounced in the remote sensing field [39]. Therefore, existing advanced disparity estimation networks are prone to significant accuracy degradation when applied to satellite images from different geographical regions or with different payloads.

### B. Domain Adaptation of Disparity Estimation Method

To reduce the performance decline of end-to-end disparity estimation networks in new scenes, many researchers have conducted cross-domain generalization studies. Existing domain adaptation methods for disparity estimation networks can be broadly divided into three categories: adversarial training, self-training, and auxiliary data training.

*1) Adversarial Training:* Zhang et al. [40] employ discriminators and unsupervised mixed training to conduct explicit domain-adversarial training and implicit feature alignment, respectively. This process ultimately achieves cross-domain disparity estimation capability. HVT [41] transforms the source domain into new global-, local-, and pixel-level domains to maximize visual differences between domains. On this basis, the feature extractor of HVT minimizes cross-domain differences to capture domain-invariant features related to disparity estimation.

*2) Self-Training:* Shen et al. [38] conducted an adaptive disparity search space at each level within network based on disparity uncertainty estimation. This method combines confidence-based pseudo-supervision for cross-domain alignment. Rao et al. [42] introduce a masked representation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

learning strategy for dense stereo matching tasks to randomly mask and recover partial information in the left-view image. The dual-task architecture was executed in parallel to enhance the generalization performance of disparity estimation network. AdaStereo [43] employs a nonadversarial progressive color transfer method and a parameter-free cost normalization layer for image- and FL domain alignment, respectively. Additionally, it incorporates a self-supervised occlusion-aware reconstruction auxiliary task to further narrow the gap between the domains.

*3) Auxiliary Data Training:* Poggi et al. [44] used sparse depth values obtained from light detection and ranging (LiDAR) as a reference and employed Gaussian functions to activate relevant features and suppress irrelevant features. Based on the same principle, EGDepth [45] included sparse disparity values for disparity completion and feature fusion and applied the obtained semi-dense disparity to variably weighted Gaussian guidance of the 4-D cost volume. EGDepth obviously enhances the robustness of disparity estimation in new domains. Huang et al. [46] employed a sparse signal super-resolution technique to extend sparse depth values and used confidence guidance from different regions to conduct end-to-end cross-domain adaptive training. GraftNet [47] integrated a general feature representation layer pretrained on large-scale data with a cosine similarity cost measure grafted into the cost aggregation module. All the above methods effectively improved the cross-domain robustness of stereo matching.

In the fields of photogrammetry and remote sensing, Xia et al. [48] trained on aerial data using GANet [14] and fine-tuned on satellite imagery, preliminarily demonstrating the transferability of disparity estimation networks to satellite images. Wu et al. [25] extensively validated the transferability of learning-based disparity estimation algorithms for geospatial applications.

Existing disparity estimation domain adaptation methods have been designed for single networks without considering their plug-and-play capabilities. Furthermore, all the aforementioned domain adaptation methods have only been validated in the field of computer vision, and their feasibility for OSSIDE remains to be verified.

## III. Proposed Method

### A. Problem Overview

This study aimed to enhance the cross-domain generalization capability of the OSSIDE task. The task involves a disparity estimation dataset as the source domain $M_S$. This source domain consists of satellite stereo image pairs $\{L_i^s, R_i^s\}_{i=1,n\in\mathbb{R}}^n$ and the corresponding dense disparity ground truth $\{D_i\}_{i=1,n\in\mathbb{R}}^n$. Additionally, the target domain $M_t$ is a disparity estimation dataset consisting solely of satellite stereo image pairs $\{L_j^t, R_j^t\}_{j=1,m\in\mathbb{R}}^m$, which have data distribution (resolution, radiation characteristics, spectral brightness, stereo imaging modes, and so on) differences from the source domain. The objective of this task is to train a cross-domain generalized OSSIDE network based on the source domain $M_S$ and enable the network to have accurate disparity estimation in the target domain $M_t$.

### B. Proposed Method

To mitigate performance degradation in end-to-end disparity estimation networks when applied to cross-domain data, this study introduces a hierarchical domain adaptation framework for end-to-end OSSIDE methods, termed HDADE. The overall architecture of HDADE is shown in Fig. 2. Specifically, the HDADE requires three core components: a disparity estimation network, source-domain data $M_s = \{L_i^s, R_i^s, D_i\}_{i=1,n\in\mathbb{R}}^n$, and target-domain data $M_t = \{L_j^t, R_j^t\}_{j=1,m\in\mathbb{R}}^m$. HDADE implements multiple technical modules to enhance the disparity estimation applicability of network on the target domain. First, source-domain data $M_s$ undergo quality evaluation using existing sparse matching techniques to select high-quality data pairs $\widehat{M}_s$. The optimized source data $\widehat{M}_s$ undergo coordinated data augmentation to generate extended data $\widehat{M}_s^D$. Subsequently, HDADE quantitatively analyzes the spectral and disparity distribution characteristics of the data $\widehat{M}_s^D$ and target data $M_t$. Based on the identified distribution differences, $\widehat{M}_s^D$ is adaptively processed to achieve explicit alignment with $M_t$ through spectral-disparity distribution fitting. Next, the fit source data $\widehat{M}_s^{DI}$ and target data $M_t$ are sequentially fed into the disparity estimation network. The 2-D/3-D domain discriminators are employed to reduce cross-domain feature sensitivity during joint training. Finally, a robust disparity estimation network and sparse matching priors are combined to generate pseudo-labels. These labels undergo refinement and facilitate direct feature alignment through self-supervision based on confidence.

*1) DL Data Optimization and Augmentation:* The scarcity of LiDAR data has led to the limited scale of the OSSIDE dataset, which utilizes LiDAR-derived ground-truth annotations. This constrained data volume significantly impedes the learning of domain-invariant representations. In addition, there are significant temporal discrepancies between the stereo satellite image pairs and LiDAR-derived ground truth [26]. These discrepancies can lead to inconsistencies in the training data pairs, thereby affecting the performance of OSSIDE networks [35]. In cross-domain generalization tasks, limited data scale and quality synergistically degrade model performance, resulting in unpredictable disparity estimation errors in the target domain. Therefore, it is necessary to evaluate the quality of each data pair, perform dataset optimization, and implement augmentation strategies on the high-quality data pairs.

Based on previous studies [35], this article employs the DL-DOA technique to enhance both data quality and volumetric sufficiency. The DL-DOA architecture incorporates two technical subcomponents: sparse matching-guided data evaluation and optimization (SMG-DEO) and stereo image pair dual augmentation (SIPDA) (as shown in Fig. 3).

*a) SMG data evaluation and optimization:* To perform the quality evaluation of data pairs, we designed the SMG-DEO. Specifically, sparse matching point pairs $P_s = \{(x_i^L, y_i^L), (x_i^R, y_i^R)\}_{i=1,k\in\mathbb{R}}^k$ are first obtained using feature detection and matching algorithms [49], [50] from the source domain $M_s$. The matching points are then refined using ratio purification and epipolar constraint [35] to obtain the optimal sparse matching points $\hat{P}_s = \{(\hat{x}_i^L, \hat{y}_i^L), (\hat{x}_i^R, \hat{y}_i^R)\}_{i=1,\hat{k}\in\mathbb{R}}^{\hat{k}}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

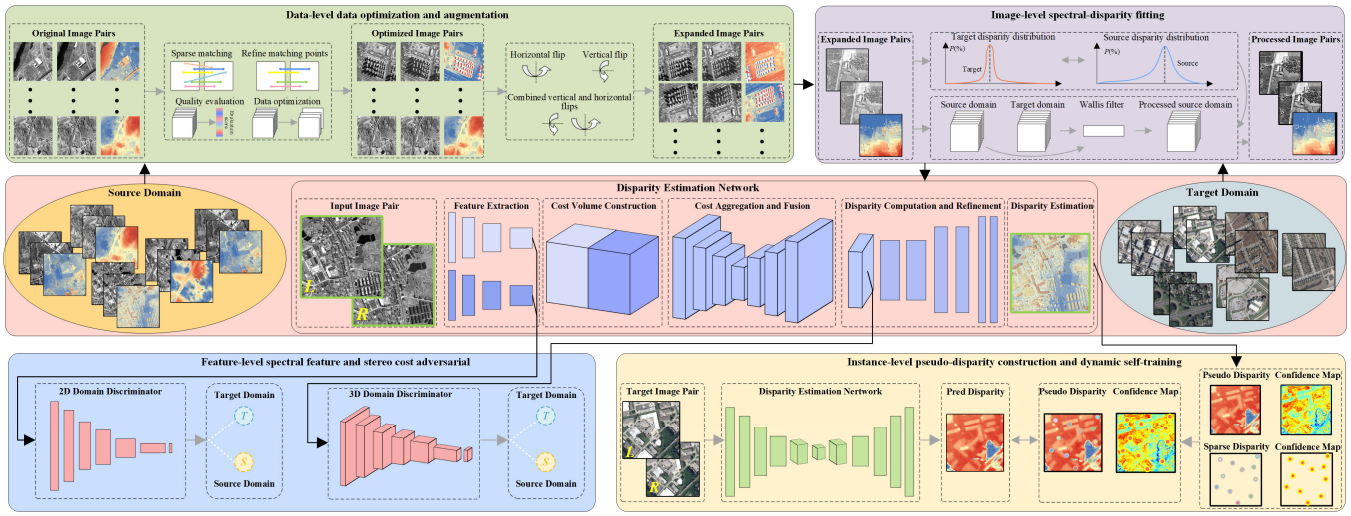



Fig. 2. Overall framework of HDADE.

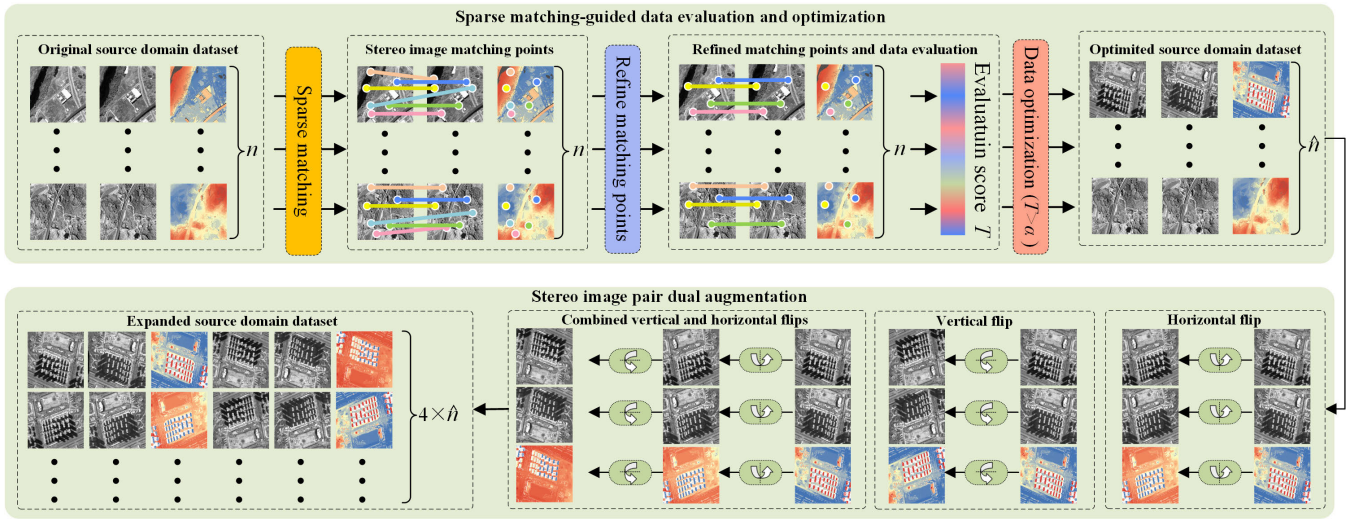



Fig. 3. DL-DOA.

Finally, the sparse disparity values of the refined sparse matching points are calculated. The absolute differences between sparse disparity values and the corresponding regions of the dense disparity ground truth are weighted and averaged to obtain a reliable quality evaluation score $T$ for the corresponding data pairs

$$M_s = \left\{L_i^s, R_i^s, D_i, T_i\right\}_{i=1}^{n}, T_i = \frac{\sum_{j=0}^{\hat{k}} \left|\left(\hat{x}_j^L - \hat{x}_j^R\right) - D_i\left(\hat{x}_j^L, \hat{y}_j^L\right)\right|}{\hat{k}} \tag{1}$$

where $D(x, y)$ is the disparity value of ground truth corresponding to the sparse matching point position $(x, y)$ in the left-view image. To exclude data pairs with significant content differences and avoid introducing errors during training that could negatively impact the network generalization ability, this study uses a data evaluation score threshold $\alpha$ to select the data for training. The source-domain data participating in disparity estimation network training can be defined as follows:

$$\widehat{M}_s = \left\{L_i^s, R_i^s, D_i, T_i\right\}_{i=1}^{\hat{n}}, T_i > \alpha. \tag{2}$$

*b) Stereo image pair dual augmentation:* To further enhance the diversity of stereo matching data, the image pairs in $\widehat{M}_s$ will undergo horizontal, vertical, and horizontal–vertical duality flips to achieve SIPDA

$$\widehat{M}_s^{FV} = \left\{F^v\left(L_i^s\right), F^v\left(R_i^s\right), F^v(D_i), T_i\right\}_{i=1}^{\hat{n}} \tag{3}$$

$$\widehat{M}_s^{FH} = \left\{F^h\left(L_i^s,\right), F^h\left(R_i^s\right), -F^h(D_i), T_i\right\}_{i=1}^{\hat{n}} \tag{4}$$

$$\widehat{M}_s^{FVH} = \left\{F^{vh}\left(L_i^s,\right), F^{vh}\left(R_i^s\right), -F^{vh}(D_i), T_i\right\}_{i=1}^{\hat{n}} \tag{5}$$

where $F^v$, $F^h$, and $F^{vh}$ are the vertical, horizontal, and vertical–horizontal flip functions, respectively. After the aforementioned process, the DL expanded dataset $\widehat{M}_s^D$ is obtained

$$\widehat{M}_s^D = \left\{\widehat{M}_s, \widehat{M}_s^{FV}, \widehat{M}_s^{FH}, \widehat{M}_s^{FVH}\right\}. \tag{6}$$

Thus, DL-DOA can effectively detect and eliminate low-quality stereo matching data pairs that may mislead the network. Furthermore, it performs dual augmentation on high-quality data pairs to enhance data diversity. The DL-DOA ensures the fitting quality of the OSSIDE network training

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

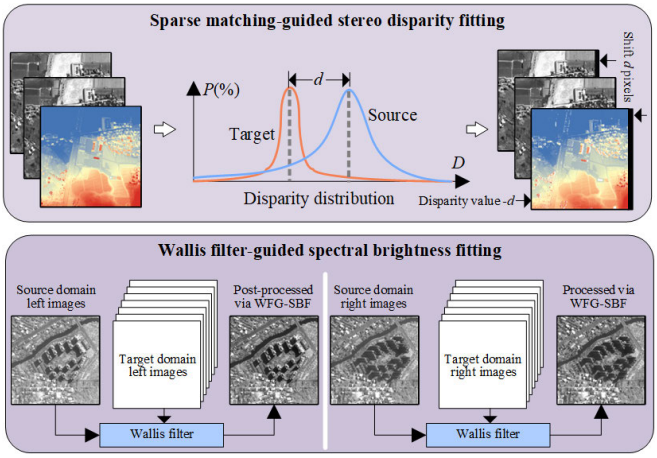



Fig. 4. IL-SDF.

and avoids amplifying the disparity estimation errors in cross-domain tasks.

*2) Image-Level Spectral-Disparity Fitting:* For image-level cross-domain tasks, end-to-end approaches suffer from performance degradation due to radiometric discrepancies. Previous studies have focused on using learning-based methods to reduce the spectral tone and brightness differences between the source and target domains [43], [51]. However, the learning-based radiometric correction requires extensive paired data, which is unavailable in cross-domain scenarios [52], [53]. Additionally, cross-domain OSSIDE tasks also face a critical challenge: disparity distribution shifts caused by stereo imaging characteristics. However, previous OSSIDE methods often neglect to bridge these gaps due to their reliance on homogeneous data assumptions.

In this study, we propose an image-level spectral-disparity fitting (IL-SDF) technique for comprehensive image-level cross-domain adaptation. IL-SDF integrates SMG stereo disparity fitting (SMG-SDF) and Wallis filter-guided spectral brightness fitting (WFG-SBF) (as shown in Fig. 4).

*a) SMG stereo disparity fitting:* To achieve stereo disparity distribution alignment in cross-domain OSSIDE data, this study employs the SMG-SDF technique. SMG-SDF is a physics-aware adaptation mechanism that leverages sparse tie points as cross-domain disparity distribution anchors. The process consists of the following steps.

First, the sparse matching method described in Section III-B1.a is used to obtain the sparse matching point set of the target-domain stereo pairs $P_t = \{(x_i^L, y_i^L), (x_i^R, y_i^R)\}_{i=1,h\in\mathbb{R}}^{h}$. Next, the corresponding sparse disparity $S$ is calculated based on the sparse matching points. Finally, the sparse disparity $S$ is encoded in the target domain

$$M_t = \left\{L_i^t, R_i^t, S_i\right\}_{i=1}^{m}, S_{i\left(x_j^L, y_j^L\right)} = \left(x_j^L - x_j^R\right) \tag{7}$$

where $S_{(x^L, y^L)}$ denotes the disparity values calculated from the sparse matching points located at $(x^L, y^L)$ in the left image and their corresponding matching points $(x^R, y^R)$ in the right image.

Second, assuming that the disparity distribution range of the source domain is $[d_{\min}^s, d_{\max}^s)$, the disparity distribution range of the target domain is determined by the following steps:

$$R^s = d_{\max}^s - d_{\min}^s \tag{8}$$

$$C\left(d_{\min}^t, d_{\max}^t, R^s, \beta\right) = \left\{\sum_{d=d_{\min}^t}^{d_{\min}^t+R^s} p(d), \sum_{d=d_{\min}^t+\beta}^{d_{\min}^t+R^s+\beta} p(d), \ldots, \sum_{d=d_{\max}^t-R^s}^{D_{\max}^t} p(d)\right\} \tag{9}$$

$$\hat{d}_{\min}^t, \hat{d}_{\max}^t = \max\left(C\left(d_{\min}^t, d_{\max}^t, R^s, \beta\right)\right) \tag{10}$$

where $R^s$ represents the disparity search range length of the source domain; $p(d)$ denotes the distribution frequency of the disparity data at a certain disparity $d$; $C(d_{\min}^t, d_{\max}^t, R^s, \beta)$ is the cumulative distribution frequency set in $[d_{\min}^t, d_{\max}^t)$ with a sampling range of $R^s$ and a step size of $\beta$; $[d_{\min}^t, d_{\max}^t)$ is the initial disparity range of the target domain, in which it is necessary to ensure that $[d_{\min}^t, d_{\max}^t) \ni [d_{\min}^s, d_{\max}^s)$; $d_{\min}^t$ and $d_{\max}^t$ must be predefined as integer multiples of $\beta$; and $\max(\cdot)$ is used to obtain the starting and ending disparity values of the maximum element in the cumulative disparity distribution set $C$. The final disparity range of the target domain is $[\hat{d}_{\min}^t, \hat{d}_{\max}^t)$.

Third, the approximate disparity distribution of the two domains can be derived by calculating the subinterval cumulative disparity distribution frequency of the target-domain sparse disparity $S$ and the source-domain dense disparity ground truth $D$

$$d_{o\min}, d_{o\max} = \max\left(C\left(d_{\min}, d_{\max}, \frac{R^s}{\tau}, 1\right)\right) \tag{11}$$

$$d_{o\mathrm{med}} = \frac{(d_{o\min} + d_{o\max})}{2} \tag{12}$$

$$\hat{d}_{o\mathrm{med}}^t = \hat{d}_{\min}^t + d_{o\mathrm{med}}^s - d_{\min}^s \tag{13}$$

where $d_{o\min}$ and $d_{o\max}$ represent the minimum and maximum values of the main disparity distribution interval, respectively; $\tau$ is the number of subintervals; $d_{o\mathrm{med}}$ is the disparity median of the primary disparity distribution interval; and $\hat{d}_{o\mathrm{med}}^t$ is the virtual disparity distribution peak of the source-domain data mapped to the target domain.

Finally, to enhance the similarity in the disparity distribution between the source and target domain, the source-domain data $\hat{M}_s$ will undergo stereo disparity shift fitting

$$\hat{M}_s^{DS} = \left\{\left[\overleftarrow{L}_i^s\left(d_{o\mathrm{med}}^t - \hat{d}_{o\mathrm{med}}^t\right), R_i^s\right] || \left[L_i^s, \vec{R}_i^s\left(d_{o\mathrm{med}}^t - \hat{d}_{o\mathrm{med}}^t\right)\right] D_i - \left(d_{o\mathrm{med}}^t - \hat{d}_{o\mathrm{med}}^t\right), T_i\right\}_{i=1}^{\hat{n}} \tag{14}$$

where $L^s$, $R^s$, $D_i$, and $T_i$ represent the left images, right images, disparity ground truth, and quality evaluation score of the source domain, respectively. $\hat{n}$ is the number of data pairs of source domain, $d_{o\mathrm{med}}^t$ is the disparity median of the primary disparity distribution interval of the target domain, $\overleftarrow{I}(a)$ represents the image $I$ shifted to the left by $a$ pixels, $\vec{I}(a)$ indicates the image $I$ shifted to the right by $a$ pixels, and $A||B$ represents the OR operation between functions $A$ and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

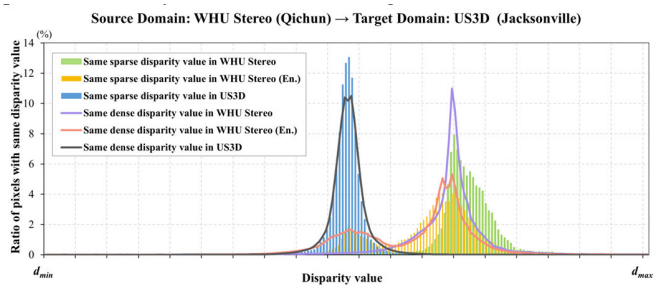



Fig. 5. Distribution of dense and sparse disparity features before and after processing with SMG-SDF (with the transfer of WHU_Stereo-Qichun to US3D-Jacksonville dataset as an example). "En." denotes the dataset processed with the SMG-SDF techniques.

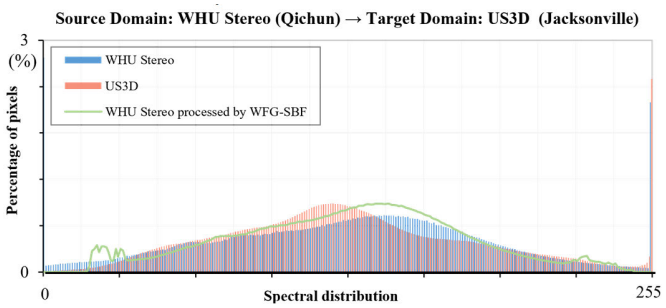



Fig. 6. Distribution of spectral brightness before and after processing with WFG-SBF (with the transfer of WHU_Stereo-Qichun to US3D-Jacksonville dataset as an example).

$B$. After the aforementioned process, $\widehat{M}_s^D$ has been expanded to an image-level processing dataset $\widehat{M}_s^{DI}$

$$\widehat{M}_s^{DI} = \left\{ \widehat{M}_s, \widehat{M}_s^{DS}, \widehat{M}_s^{FV}, \widehat{M}_s^{FH}, \widehat{M}_s^{FVH} \right\}. \tag{15}$$

As illustrated in Fig. 5, after processing with the SMG-SDF, the source-domain OSSIDE data can establish a disparity distribution peak that is nearly consistent with the target domain.

*b) Wallis filter-guided spectral brightness fitting:* The Wallis filter can dynamically adjust pixel intensities within sliding windows to match target-domain statistical characteristics and does not require extensive training data. We implement the WFG-SBF to effectively conduct spectral brightness distribution alignment toward the target domain.

Based on the image $I^s$ from $\widehat{M}_s^D$, the processing steps of WFG-SBF are as follows:

$$\hat{I}^s_{(x,y)} = \left(I^s_{(x,y)} - \mu^s\right) \frac{\sigma^t}{\sigma^s + \epsilon} + \mu^t \tag{16}$$

where $I_{(x,y)}$ denotes a pixel located at the position $(x, y)$ in the image; $\mu$ and $\sigma$ represent the mean and variance of the grayscale values of pixels within the local region surrounding the pixel $(x, y)$, respectively; and $\epsilon$ denotes the smoothing coefficient. The WFG-SBF aligns the spectral brightness of stereo images from the source-domain dataset with the target domain. After being processed by WFG-SBF, the final version of image-level processing dataset $\widehat{M}_s^{DI}$ is obtained

$$\widehat{M}_s^{DI} = \left\{ \overline{\widehat{M}_s}, \overline{\widehat{M}_s^{DS}}, \overline{\widehat{M}_s^{FV}}, \overline{\widehat{M}_s^{FH}}, \overline{\widehat{M}_s^{FVH}} \right\} \tag{17}$$

where $\overline{M}$ represents that the data have been processed by WFG-SBF.

As illustrated in Fig. 6, the WFG-SBF processed source-domain data achieves partial spectral distribution alignment with the target domain, demonstrating significant improvement in cross-domain radiometric consistency.

IL-SDF explicitly aligns the data distribution characteristics of the source-domain stereo image pairs with those of the target domain, thereby reducing the risk of disparity estimation network accuracy loss in the target domain.

*3) FL Spectral Feature and Stereo Cost Adversarial:* IL-SDF aligns the source-domain OSSIDE data with the target-domain data regarding spectral brightness and stereo disparity distribution at the image level. However, this process does not account for the learning of cross-domain invariant knowledge within the network. When the target-domain data distribution undergoes partial changes or fluctuations, the network remains susceptible to producing high-entropy disparity estimation results. Therefore, a critical challenge is to promote networks that reveal cross-domain knowledge distribution characteristics and learn general cross-domain invariance features.

This study proposes an FL-SSAdv method to address the aforementioned issues. Unlike previous domain adaptation methods, FL-SSAdv reasonably integrates a 2-D domain discriminator based on spectral features and a 3-D domain discriminator based on stereo matching cost into the OSSIDE network. During the representation learning, it incorporates 2-D and 3-D adversarial losses to implicitly address the data distribution differences in high dimensions across multiple granularity levels within the network. This ultimately achieves adaptive cross-domain fine-grained feature sensitivity reduction within the OSSIDE network.

As shown in Fig. 7, FL-SSAdv comprises three networks (OSSIDE network $N$, 2-D domain discriminator network $N_{2D}$, and 3-D domain discriminator network $N_{3D}$), along with a cross-domain disparity transformation layer. The OSSIDE network $N$ is trained on the disparity estimation dataset $\widehat{M}_s^{DI}$ using the smooth $L1$ loss function to obtain disparity estimation loss $\text{Loss}_{\text{DE}}$

$$\text{Loss}_{\text{DE}} = \text{Smooth}L1(P_{\text{DE}}, D) \tag{18}$$

where $P_{\text{DE}}$ is the prediction of the disparity estimation network and Smooth$L1$ represents the smooth L1 loss function. Notably, the HDADE exhibits plug-and-play versatility, allowing any OSSIDE network to be effectively integrated into the proposed domain generalization framework.

In the FL-SSAdv training stage, the network $N$ processes both source-domain data $\widehat{M}_s^{DI}$ and target-domain data $M_t$, dynamically adjusting disparity ranges between $[d^s_{\min}, d^s_{\max})$ and $[\hat{d}^t_{\min}, \hat{d}^t_{\max})$ based on the disparity transformation layer. This dual-domain processing enables the network $N_{2D}$ to extract the 2-D feature map from the feature extraction layer, while the network $N_{3D}$ decodes the 3-D cost volume from the cost aggregation module. Then, networks $N_{2D}$ and $N_{3D}$ are used to perform domain classification. Predictions $P_{2D}$ and $P_{3D}$ of the two domain discriminators are utilized to compute the domain discrimination losses $\text{Loss}_{\text{DD}-2D}$ and $\text{Loss}_{\text{DD}-3D}$ based

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

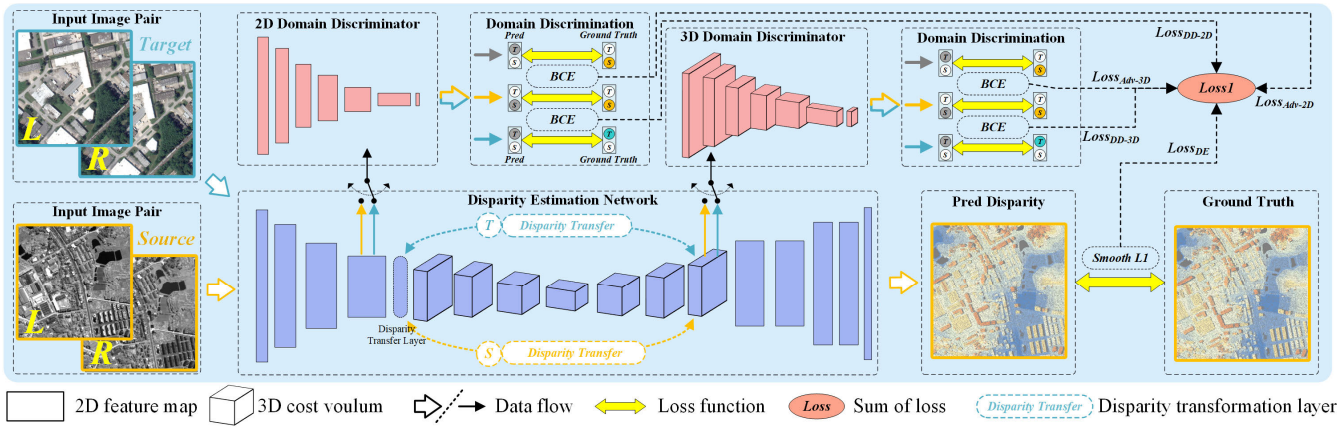



Fig. 7. FL-SSAdv.

on the domain ground truths $D_{2D}$ and $D_{3D}$, along with the binary cross-entropy loss function. The domain discrimination loss $\text{Loss}_{DD}$ can be defined as follows:

$$\text{Loss}_{DD} = -\left(\frac{1}{h \times w}\right) \sum_{i=1}^{h} \sum_{j=1}^{w} D_{(i,j)} \ln\left(P_{(i,j)}\right) + \left(1\text{-}D_{(i,j)}\right) \ln\left(1 - P_{(i,j)}\right) \quad (19)$$

where $h$ and $w$ represent the height and width of the feature map output by the domain discriminator networks, respectively. Additionally, during the supervised learning of the disparity estimation and domain discrimination networks, adversarial losses $\text{Loss}_{Adv-2D}$ and $\text{Loss}_{Adv-3D}$ are employed to encourage the internal parameters of network $N$ in updating the gradients toward a direction that confuses $N_{2D}$ and $N_{3D}$. The adversarial losses are formulated as follows:

$$\text{Loss}_{Adv} = -\left(\frac{1}{h \times w}\right) \sum_{i=1}^{h} \sum_{j=1}^{w} \left(\check{D}_{(i,j)}\right) \ln\left(P_{(i,j)}\right) + \left(1 - \check{D}_{(i,j)}\right) \ln\left(1 - P_{(i,j)}\right), \check{D} = -D + 1 \quad (20)$$

where $\check{D}$ represents the inverted domain ground truth used to mislead the two domain discriminators $N_{2D}$ and $N_{3D}$. At this stage, FL-SSAdv participates in an ongoing adversarial game centered on visual features and stereo costs. $N_{2D}$ and $N_{3D}$ strive to enhance their domain discrimination capabilities of feature map and stereo cost volume in $N$, while $N$ aims to achieve domain insensitivity in visual feature extraction and stereo matching cost aggregation to confuse $N_{2D}$ and $N_{3D}$. Finally, the domain discriminator $N_{2D}$ encourages the disparity estimation network $N$ to reduce the sensitivity to spectral brightness changes between the source and target domains. The domain discriminator $N_{3D}$, promotes the robustness of the disparity estimation network against variations in stereo disparity distribution between domains. The overall loss during this stage is defined as follows:

$$\text{Loss1} = \text{Loss}_{DE} + \gamma(\text{Loss}_{DD-2D} + \text{Loss}_{Adv-2D} + \text{Loss}_{DD-3D} + \text{Loss}_{Adv-3D}) \quad (21)$$

where $\gamma$ is the weight for the domain discriminator loss. As described above, the domain discriminator networks $N_{2D}$ and $N_{3D}$ engage in fine-grained adversarial training across 2-D and 3-D feature spaces. Ultimately, FL-SSAdv ensures that network $N$ possesses the invariance capability for cross-domain spectral feature capture and stereo cost aggregation.

*4) IL Pseudo-Disparity Construction and Dynamic Self-Training:* The aforementioned approaches only indirectly guide the disparity estimation network to learn the target domain. Therefore, we construct an instance-level pseudo-disparity construction and dynamic self-training (IL-PCDS) based on target-domain instance data (as shown in Fig. 8). The IL-PCDS enables the disparity estimation network to directly establish a stereo matching paradigm aligned with actual data distribution of the target domain.

Initially, the IL-PCDS loads the disparity estimation network weights obtained from the FL-SSAdv to perform disparity estimation for the target domain. Next, to further ensure the unbiased nature of the pseudo-disparity, IL-PCDS combines the target-domain sparse disparity $S$ obtained from (7) with the disparity predictions from FL-SSAdv, forming a sparse and dense multisource pseudo-disparity output $\overline{D}$

$$\overline{D} = N_{\text{FL-SSAdv}}\left(L^t, R^t\right) \cup S \quad (22)$$

where $N_{\text{FL-SSAdv}}$ is the disparity estimation network obtained from the FL-SSAdv. The multisource target-domain pseudo-disparity map $\overline{D}$ provides more direct guidance for the network to capture task-related knowledge on real target-domain data. However, errors in the pseudo-disparity during self-training may introduce optimization biases into the network, potentially compromising the network's stability and reliability. Therefore, this study introduces a multisource confidence map to adaptively optimize the pseudo-supervised stage of the network.

Specifically, existing disparity estimation networks frequently use a soft-argmin operator to transform the cost volume into continuous disparity expectations. Then, the disparity expectations are obtained through a probability-weighted summation operation incorporating subpixel disparity. However, the cost curves in dense matching often exhibit a multipeak distribution. Previous works indicate that disparity estimation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

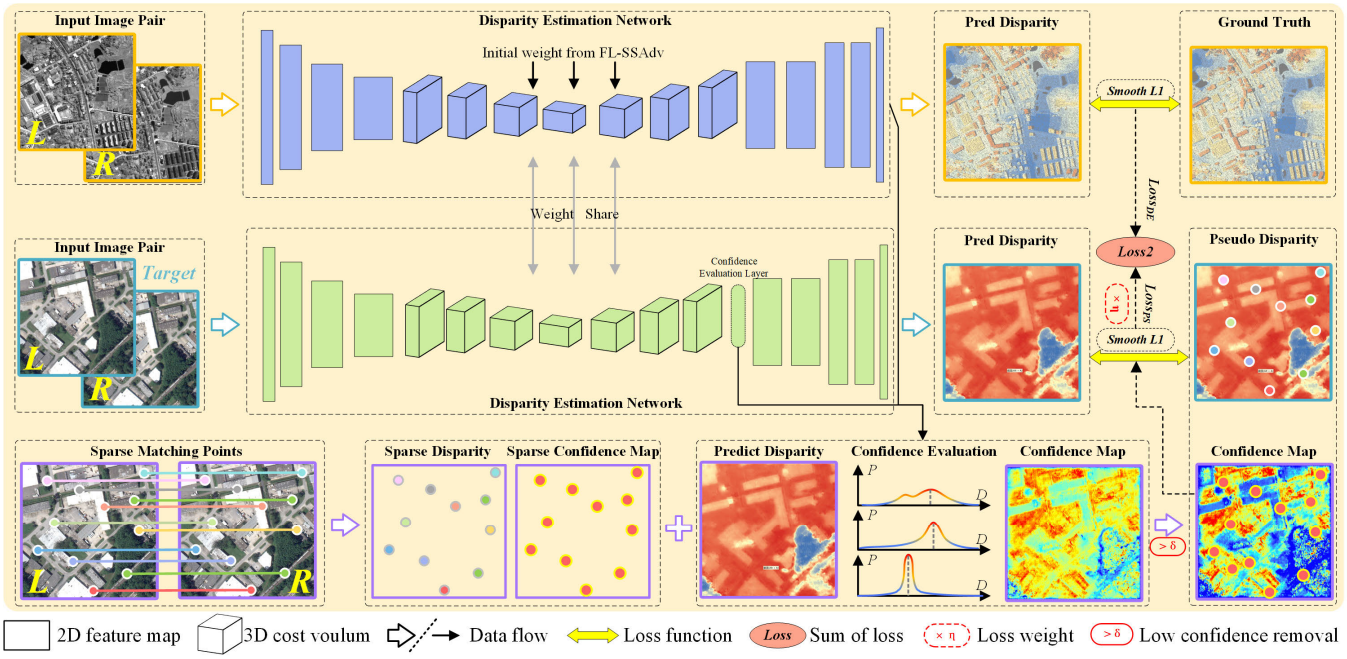



Fig. 8. IL-PCDS.

errors are positively correlated with the degree of multipeak distribution [54]. Following this principle, we designed a reliability measurement method for the sparse-dense multisource pseudo-disparity $\bar{D}$:

$$\bar{E} = \left\lfloor \sum_{i=d_{\min}}^{d_{\max}} \left( \frac{e^{V_i}}{\sum_{j=d_{\min}}^{d_{\max}} e^{V_j}} \right)^2 \right\rfloor_{\delta} \cup (S \times \rho) \tag{23}$$

where $\lfloor A \rfloor_{\delta}$ sets the values in matrix $A$ to 0 if they are less than the confidence threshold $\delta$; $V$ is the cost volume inside the stereo matching network; $d_{\max}$ and $d_{\min}$ are the maximum and minimum disparity, respectively; $\rho$ is the sparse disparity confidence; and $\bar{E}$ is the constructed multisource confidence map. The pseudo-supervised loss $\text{Loss}_{\text{PS}}$ and overall loss in IL-PCDS can be defined as follows:

$$\text{Loss}_{\text{PS}} = \text{Smooth}L1\left(P_{\text{DE}}, \bar{D}\right) \cdot \bar{E} \tag{24}$$

$$\text{Loss2} = \text{Loss}_{\text{DE}} + \eta(\text{Loss}_{\text{PS}}) \tag{25}$$

where $\eta$ is the weight for the pseudo-supervision loss. The IL-PCDS strategy can directly assist the disparity estimation network in adapting to the real data distribution characteristics of the target domain. Meanwhile, IL-PCDS assists our method in constructing a comprehensive cross-domain feature alignment pipeline across scales, spectral, and stereo characteristics.

## IV. Experiment Setting

In this section, we first provide an overview of the experimental data. Next, we describe the evaluation metrics used in this study. Finally, we provide details on the experimental environment and the implementation of the methods.

### A. Datasets

In this experiment, we used two representative publicly available OSSIDE datasets, WHU_Stereo [26] and US3D [55], to evaluate the domain generalization performance of the experimental methods. Additionally, we collected and constructed large-scale GF7 stereo images with various types of terrain worldwide to qualitatively validate the experimental methods.

WHU_Stereo is a highly challenging OSSIDE dataset. The high-resolution stereo satellite image pairs in the dataset were captured by the dual-line array stereo camera on the GaoFen-7 (GF-7) satellite. The GF-7 satellite is equipped with forward and backward stereo mapping cameras that provide near-synchronous stereo satellite images with a spatial resolution better than 0.8 m. The disparity data are obtained by airborne LiDAR point clouds. WHU_Stereo contains 1757 data pairs, covering the cities of Shaoguan, Kunming, Yingde, Qichun, Wuhan, and Hengyang.

US3D is a large-scale OSSIDE dataset with semantic annotations. The dataset includes epipolar-rectified stereo image pairs from the WorldView-3 satellite and 3-D ground truth constructed from airborne LiDAR point clouds. The ground sampling distance of the image data is approximately 35 cm, covering Jacksonville and Omaha from 2014 to 2016.

To test the effectiveness of traditional, end-to-end, and domain adaptation methods for disparity estimation across diverse terrains, architectural styles, and large-scale scenes, this study collected GF-7 stereo data from five globally distributed regions and constructed a dataset called LargeS-GF7. LargeS-GF7 includes various terrains such as deserts, mountains, forests, villages, cities, and farmlands, covering parts of the Americas, Europe, Oceania, and Asia. LargeS-GF7 enables a more comprehensive visual evaluation of the generalization performance for the methods discussed in this study.

To evaluate the generalizability of the methods for cross-spatiotemporal data and cross-sensor data, WHU_Stereo

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

and US3D were divided based on the regional location of the images. As shown in Fig. 9, WHU_Stereo-Qichun and US3D-Omaha data are set as the source domains, while WHU_Stereo-5cities (Hengyang, Shaoguan, Yingde, Kunming, and Wuhan) and US3D-Jacksonville data are set as the target domains. In the experiments, training will be conducted based on the source-domain data, and the performance of methods will be tested on the target-domain test sets. Additionally, to ensure the reliability of the test set evaluation results, the evaluation score threshold $\alpha$ for all test set data is set to 3.

### B. Evaluation Metrics

To quantitatively describe the reliability of the methods, we used the average endpoint error (EPE) and the fraction of erroneous pixels ($D1$) to evaluate the disparity estimation results in combination with the ground truth.

1) *EPE:* It represents the absolute average $L1$ distance between the prediction and ground truth in the valid region (the lower, the better)

$$\mathrm{EPE} = \frac{1}{n}\sum_{i=1}^{n} |P_{\mathrm{DE}_i} - D_i|. \tag{26}$$

where $n$ is the number of valid grid units in the ground truth, $P_{\mathrm{DE}_i}$ is the predicted disparity value at the $i$th grid unit, and $T_i$ is the ground-truth disparity at the $i$th grid unit.

2) *RMSE:* It is used to represent the standard deviation of the difference between the ground truth and estimated disparity in the valid region (the lower, the better)

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(P_{\mathrm{DE}_i} - D_i\right)^2}. \tag{27}$$

3) *D1:* It represents the ratio of the number of estimated disparity values whose absolute error is within a threshold to the number of valid grid units (the lower, the better)

$$D1 = \frac{\sum_{i=1}^{n} I\left(|P_{\mathrm{DE}_i} - D_i| < \varepsilon\right)}{n} \times 100\% \tag{28}$$

where $\varepsilon$ is the error threshold, set to 3 in this article, and $I(\cdot)$ is the Iverson bracket, which takes the value 1 when the condition inside is satisfied and 0 when it is not.

### C. Training Details

The experiments involving traditional methods and HMSM-Net in this study were conducted on a workstation equipped with an NVIDIA GeForce RTX 4070 12-GB GPU and an Intel[1] Core[2] i5-13600KF CPU. The experiments related to STTR were conducted on a computer workstation equipped with an NVIDIA GeForce RTX 4090 24-GB GPU and AMD EPYC 7T83 CPU. The implementation of traditional disparity estimation methods, as well as the training and testing processes of deep learning disparity estimation methods, was carried out on the Ubuntu 20.04 operating system using the OpenCV 3.4.16 and PyTorch 2.0.0 libraries.

In the experiments, traditional disparity estimation algorithms SGBM [6] and MGM [8] were selected as comparison baselines. HMSM-Net [20] and STTR [18] were used as end-to-end OSSIDE baseline networks for cross-domain (Cross-D) learning tests with source-domain training and target-domain inference, as well as self-domain (Self-D) learning tests with target-domain training and target-domain inference. In addition, end-to-end networks were also tested for generalization using domain adaptation methods from computer vision, such as domain-adversarial (D-Adv) [56] and self-ensembling (S-Ens) [57], as well as disparity estimation domain adaptation methods, such as GraftNet [47], HVT [41], EGDepth [45], and the HDADE proposed in this article. It is worth noting that GraftNet and EGDepth are challenging to integrate reasonably into the transformer-based STTR architecture. Therefore, GraftNet and EGDepth were not used in the transfer experiments based on STTR.

During the training of HDADE, the data evaluation score threshold $\alpha$ was set to 5, the number of subintervals $\tau$ was set to 14, the domain discriminator loss weight $\gamma$ was set to 0.001, the confidence threshold $\delta$ was set to 0.3, the sparse disparity confidence $\beta$ was set to 1, and the pseudo-supervision loss weight $\eta$ was set to 1.0. In the WHU_Stereo-Qichun and US3D-Omaha source domains, HDADE will be trained for 26 and 11 epochs, respectively. Among these, FL-SSAdv will train for 25 and 10 epochs, and the training of IL-PCDS will be conducted in the final epoch.

## V. Experimental Results

This section provides a comprehensive comparison of methods involved in cross-domain OSSIDE experiments from both quantitative and qualitative aspects.

### A. HMSM-Net-Based WHU-Stereo Transfer Experiment

Table I shows the accuracy comparison of disparity estimation for various methods on the WHU_Stereo-5cities as well as US3D-Jacksonville testing sets based on the WHU_Stereo-Qichun training set. Figs. 10 and 11 show the results of disparity estimation for various methods on the WHU_Stereo-5cities and US3D-Jacksonville testing sets, respectively.

The traditional disparity estimation methods, SGBM and MGM, provided relatively reliable and stable disparity estimation results for each target-domain test set. However, Figs. 10 and 11 show that SGBM and MGM have obvious disparity estimation noise and missing areas. Among the end-to-end methods based on HMSM-Net, Self-D directly performed supervised learning on the target domain and achieved the best visual effects and the highest accuracy. Owing to its lack of robustness for the target domain, Cross-D showed a significant decline in performance on both target-domain test sets, with large areas of missing disparity estimation. Compared to Self-D, Cross-D presented decreases in performance

[1]Registered trademark.
[2]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

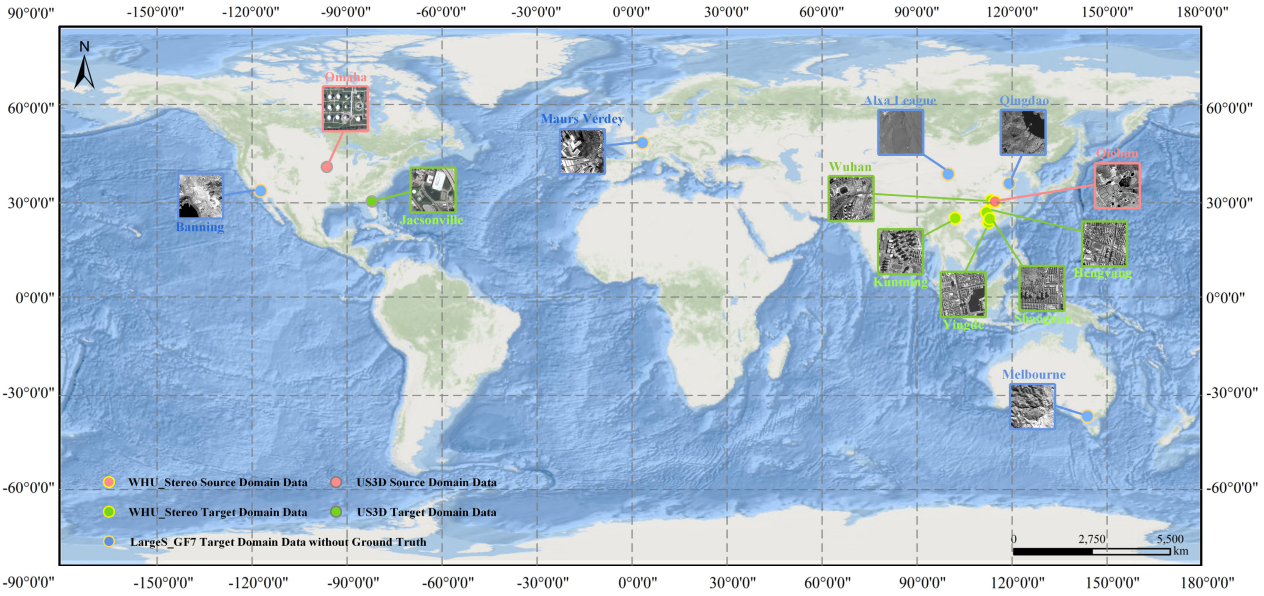



Fig. 9. Overview of the study areas.

TABLE I

QUANTITATIVE EVALUATION RESULTS OF DISPARITY ESTIMATION METHODS BASED ON WHU_STEREO-QICHUN SOURCE-DOMAIN DATA

| Dataset | Baseline | Method | 5 cities | | | Jacksonville | | |
|---|---|---|---|---|---|---|---|---|
| | | | D1 | EPE | RMSE | D1 | EPE | RMSE |
| WHU_Stereo (Qichun) | / | SGBM | 13.31% | 3.6589 | 14.6515 | 15.76% | 3.7772 | 12.3928 |
| | / | MGM | 14.48% | 2.5199 | 8.7404 | 14.10% | 2.6359 | 8.3589 |
| | HMSM-Net | Self-D | 13.13% | 1.8788 | 3.9987 | 10.84% | 1.5923 | 3.7031 |
| | | Cross-D | 24.33% | 3.0942 | 7.8911 | 26.11% | 3.5093 | 7.1743 |
| | | D-Adv | 21.35% | 2.6493 | 6.4475 | 27.49% | 3.9514 | 8.3080 |
| | | S-Ens | 19.60% | 2.5565 | 6.4588 | 24.92% | 3.3384 | 7.1894 |
| | | GraftNet | 25.32% | 3.0302 | 7.4672 | 27.89% | 2.8200 | 4.6612 |
| | | HVT | 21.44% | 2.9260 | 8.1539 | 23.35% | 3.1469 | 7.2722 |
| | | EGDepth | 22.98% | 2.9001 | 7.4853 | 24.65% | 3.7814 | 9.3748 |
| | | HDADE | 18.14% | 2.3154 | 5.1573 | 16.90% | 2.0936 | 4.0809 |

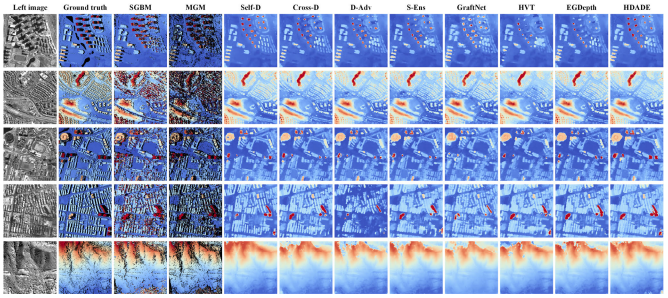



Fig. 10. Visualization of the disparity estimation results of various methods on the WHU_Stereo-5cities test set.

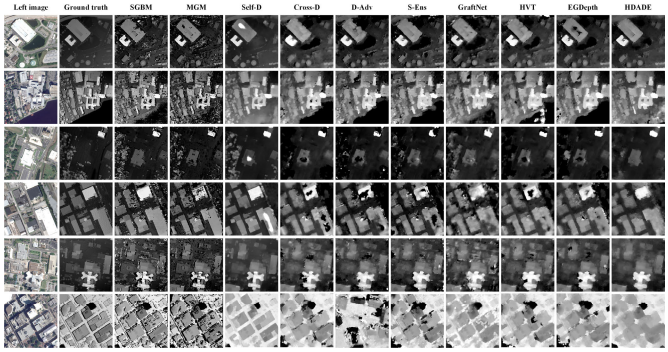



Fig. 11. Visualization of the disparity estimation results of various methods on the US3D-Jacksonville test set.

of 11.20% and 15.27% in $D1$, 1.2154 and 1.9170 in EPE, and 3.8924 and 3.4712 in RMSE for the two target-domain test sets.

Because D-Adv and S-Ens helped HMSM-Net implicitly learn target-domain knowledge in an unsupervised manner, both accuracy and visual effects were significantly improved compared to Cross-D. Among the domain adaptation methods, S-Ens achieved the second-best accuracy. GraftNet, which uses general pretrained weights for extracting low-level features, struggled to effectively extract features from satellite images, resulting in only slight improvements in accuracy and producing fragmented and blurred disparity estimation results. Both HVT and EGDepth effectively assisted HMSM-Net in reducing disparity estimation accuracy loss in the target domain; however, some missing areas remained in their disparity estimation results.

The HDADE proposed in this article helped the HMSM-Net achieve the best disparity estimation accuracy and visual effects among the domain adaptation methods. HDADE significantly reduced HMSM-Net degradation on cross-domain data. Compared to Cross-D, HDADE improved $D1$ by 6.19% and 9.21%, EPE by 0.7788 and 1.4157, and RMSE by 2.7338 and 3.0934 for the WHU_Stereo-5cities and US3D-Jacksonville test sets, respectively. Additionally, in Figs. 10 and 11, it can be seen that HDADE ensures that the disparity estimation results of HMSM-Net are smoother around ground object edges, with no large areas of disparity estimation errors or missing areas.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE EVALUATION RESULTS OF DISPARITY ESTIMATION METHODS BASED ON US3D-OMAHA SOURCE-DOMAIN DATA

| Dataset | Baseline | Method | Jacksonville | | | 5 cities | | |
|---|---|---|---|---|---|---|---|---|
| | | | D1 | EPE | RMSE | D1 | EPE | RMSE |
| US3D (Omaha) | / | SGBM | 15.76% | 3.7772 | 12.3928 | 13.31% | 3.6589 | 14.6515 |
| | / | MGM | 14.10% | 2.6359 | 8.3589 | 14.48% | 2.5199 | 8.7404 |
| | HMSM-Net | Self-D | 10.84% | 1.5923 | 3.7031 | 13.13% | 1.8788 | 3.9987 |
| | | Cross-D | 16.43% | 1.9799 | 3.6260 | 36.75% | 5.0624 | 11.5458 |
| | | D-Adv | 14.15% | 1.8153 | 3.4995 | 27.32% | 3.7916 | 9.6838 |
| | | S-Ens | 13.84% | 1.7764 | 3.4368 | 31.03% | 4.0079 | 9.8532 |
| | | GraftNet | 20.45% | 2.2961 | 4.2226 | 42.52% | 6.0096 | 12.3953 |
| | | HVT | 15.39% | 1.8803 | 3.5846 | 33.10% | 4.4472 | 10.6443 |
| | | EGDepth | 13.57% | 1.7325 | 3.3012 | 30.19% | 3.9985 | 9.2619 |
| | | HDADE | 13.52% | 1.7382 | 3.3844 | 24.96% | 2.8729 | 6.5388 |

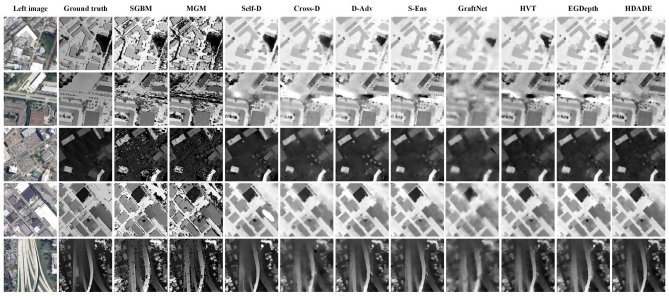



Fig. 12. Visualization of the disparity estimation results of various methods on the US3D-Jacksonville test set.

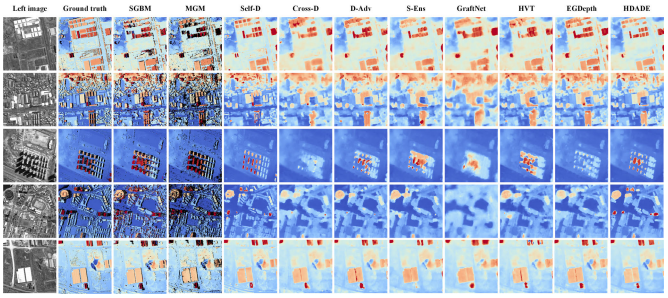



Fig. 13. Visualization of the disparity estimation results of various methods on the WHU_Stereo-5cities test set.

### B. HMSM-Net-Based US3D Transfer Experiment

Table II shows the accuracy comparison of disparity estimation for various methods trained on US3D-Omaha data and tested on the US3D-Jacksonville and WHU_Stereo-5cities test sets. Figs. 12 and 13 show the results of disparity estimation for various methods on the US3D-Jacksonville and WHU_Stereo-5cities test sets, respectively.

The results show that the traditional methods of SGBM and MGM provide relatively stable disparity accuracy but still exhibit obvious disparity prediction noise and missing areas. Except for GraftNet, all domain adaptation methods effectively reduced the performance degradation of HMSM-Net compared with Cross-D. GraftNet, which employs general low-level features from the field of computer vision, cannot easily adapt to cross-domain tasks based on the high-resolution US3D-Omaha source-domain training data. Consequently, GraftNet results in a negative optimization in the target domain.

HDADE achieved the best domain adaptation accuracy for the disparity estimation network in this experiment. Compared with Cross-D, HDADE improved the $D1$, EPE, and RMSE metrics by 2.91%, 0.2417, and 0.2416 on the Jacksonville data, and by 11.79%, 2.1895, and 5.0070 on the WHU_Stereo-5cities test set, respectively. Additionally, as shown in Figs. 12 and 13, HDADE assisted the HMSM-Net in producing excellent disparity estimation results, demonstrating significant advantages in terms of contour smoothness and fine-grained objects.

TABLE III

QUANTITATIVE EVALUATION RESULTS OF DISPARITY ESTIMATION METHODS BASED ON WHU_STEREO-QICHUN SOURCE-DOMAIN DATA

| Dataset | Baseline | Method | 5 cities | | | Jacksonville | | |
|---|---|---|---|---|---|---|---|---|
| | | | D1 | EPE | RMSE | D1 | EPE | RMSE |
| WHU_Stereo (Qichun) | / | SGBM | 13.31% | 3.6589 | 14.6515 | 15.76% | 3.7772 | 12.3928 |
| | / | MGM | 14.48% | 2.5199 | 8.7404 | 14.10% | 2.6359 | 8.3589 |
| | STTR | Self-D | 27.74% | 4.4038 | 11.2328 | 10.27% | 1.5016 | 3.3485 |
| | | Cross-D | 54.03% | 8.7399 | 17.7346 | 36.00% | 3.5232 | 6.0223 |
| | | D-Adv | 45.74% | 7.6664 | 17.6409 | 31.47% | 3.2729 | 5.8902 |
| | | S-Ens | 44.07% | 7.4039 | 16.2103 | 33.05% | 3.2944 | 5.8254 |
| | | HVT | 49.25% | 8.1622 | 16.8654 | 42.85% | 4.0771 | 6.5232 |
| | | HDADE | 35.39% | 6.0359 | 14.6604 | 27.71% | 2.9239 | 5.2862 |

### C. STTR-Based WHU-Stereo Transfer Experiment

Table III shows the quantitative accuracy comparison of disparity estimation for various methods based on WHU-Stereo-Qichun training data as well as the US3D-Jacksonville and WHU_Stereo-5cities test sets. Figs. 14 and 15 show the visual results of disparity estimation for various methods on the US3D-Jacksonville and WHU_Stereo-5cities test sets, respectively. STTR, which is based on the transformer architecture, requires a large amount of data to robustly capture features and construct a matching paradigm. However, due to the limited data volume of the WHU_Stereo-Qichun source domain, STTR experienced significant accuracy degradation in the target domain under the Cross-D, making it less reliable than SGBM and MGM. Since D-Adv and S-Ens provide limited assistance to STTR in revealing the knowledge distribution of the target domain, the improvement in evaluation accuracy and visual effects compared to Cross-D is not significant. HVT lacks generality for transformer-based networks, resulting in minimal accuracy improvement for the WHU_Stereo-5cities and even negative optimization on the US3D-Jacksonville. In this challenging experiment, HDADE also assisted STTR in achieving the highest disparity estimation accuracy improvement. Compared to Cross-D, HDADE improved the $D1$, EPE, and RMSE metrics by 18.64%, 2.7040, and 3.0742 for the WHU_Stereo-5cities test set and by 8.29%, 0.5993, and 0.7361 for the US3D-Jacksonville test set, respectively. The experimental results effectively demonstrate the generality of HDADE across different disparity estimation networks in preventing cross-domain accuracy loss.

## VI. DISCUSSION

In this section, we describe the ablation experiments conducted to functionally validate each proposed technical component. Additionally, we tested the $\alpha$-threshold impact on network performance and the usability of HDADE on large-scale, cross-temporal, and cross-terrain images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

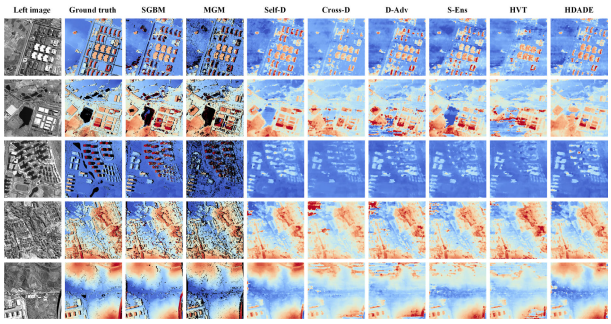



Fig. 14. Visualization of the disparity estimation results of various methods on the WHU_Stereo-5cities test set.

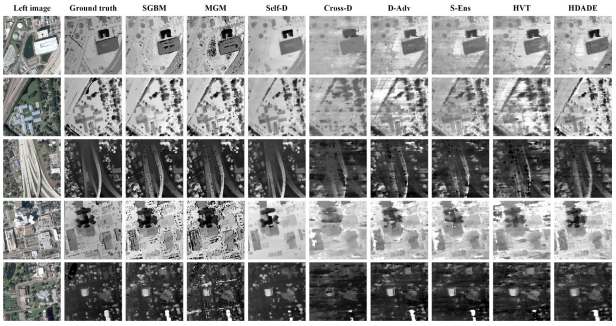



Fig. 15. Visualization of the disparity estimation results of various methods on the US3D-Jacksonville test set.

TABLE IV

QUANTITATIVE EVALUATION RESULTS OF DOMAIN TRANSFER FOR VARIOUS HDADE VARIANTS BASED ON WHU_STEREO-QICHUN SOURCE-DOMAIN DATA

| Baseline | | HMSM-Net | | | | | | |
|---|---|---|---|---|---|---|---|---|
| DL-DOA | SMG-DEO | × | √ | √ | √ | √ | √ | √ |
| | SIPDA | × | × | √ | √ | √ | √ | √ |
| IL-SSCA | SMG-SDF | × | × | × | √ | √ | √ | √ |
| | WFG-SBF | × | × | × | × | √ | √ | √ |
| FL-SSAdv | | × | × | × | × | × | √ | √ |
| IL-PCDS | | × | × | × | × | × | × | √ |
| WHU_Stereo 5 cities | D1(%) | 24.33% | 21.33% | 22.57% | 22.51% | 21.49% | 17.80% | 18.14% |
| | EPE | 3.0942 | 3.0492 | 2.9826 | 2.8459 | 2.6895 | 2.4396 | 2.3154 |
| | RMSE | 7.8911 | 8.1561 | 7.9495 | 7.0429 | 6.2483 | 5.9548 | 5.1573 |
| US3D Jacksonville | D1(%) | 26.11% | 23.92% | 20.33% | 18.77% | 18.53% | 17.89% | 16.90% |
| | EPE | 3.5093 | 2.6341 | 2.6484 | 2.2612 | 2.2518 | 2.1701 | 2.0936 |
| | RMSE | 7.1743 | 5.0281 | 5.9345 | 4.4280 | 4.3056 | 4.2592 | 4.0809 |

## A. Ablation Experiments

In this experiment, we either removed or installed various technical components of HDADE based on the HMSM-Net architecture to create a series of domain adaptation variants. These variants were trained on WHU_Stereo-Qichun source-domain data and evaluated using the WHU_Stereo-5cities and US3D-Jacksonville target domains. This setup was used to quantitatively verify and discuss the effectiveness of each technical component in enhancing the cross-domain generalization performance of disparity estimation network. The structures of multiple variant methods and their cross-domain quantitative evaluation results are presented in Table IV.

TABLE V

QUANTITATIVE EVALUATION RESULTS UNDER VARYING DATA EVALUATION SCORE THRESHOLDS BASED ON HMSM-NET AND WHU_STEREO-QICHUN SOURCE-DOMAIN DATA

| Threshold $\alpha$ | Number of training data pairs | 5 cities | | | Jacksonville | | |
|---|---|---|---|---|---|---|---|
| | | D1 | EPE | RMSE | D1 | EPE | RMSE |
| 2.0 | 277 | 27.26% | 3.5304 | 8.5935 | 26.19% | 2.6177 | 4.5050 |
| 3.0 | 398 | 24.64% | 3.1873 | 8.0137 | 25.94% | 2.6786 | 4.9524 |
| 4.0 | 451 | 22.67% | 3.1967 | 8.4344 | 24.86% | 2.7153 | 5.1971 |
| 5.0 | 479 | 21.33% | 3.0492 | 8.1561 | 23.92% | 2.6341 | 5.0281 |
| 6.0 | 500 | 21.73% | 2.8606 | 7.1081 | 24.56% | 2.6001 | 4.6836 |
| 7.0 | 524 | 21.41% | 2.8382 | 7.1906 | 24.64% | 2.6017 | 4.6427 |
| 8.0 | 553 | 26.02% | 3.2821 | 8.5482 | 27.22% | 2.6703 | 4.4189 |
| 9.0 | 573 | 22.66% | 2.9042 | 7.3528 | 25.69% | 2.6769 | 4.8086 |
| / | 600 | 24.33% | 3.0942 | 7.8911 | 26.11% | 3.5093 | 7.1743 |

According to Table IV, it is evident that each domain adaptation technical component contributes to the accuracy benefits for the disparity estimation network. Specifically, the SMG-DEO can effectively improve the data quality in the source-domain test set, resulting in an accuracy improvement. SIPDA performed dual augmentation on the data pairs in the source-domain dataset, and the significant increase in data volume led to a noticeable improvement in the accuracy.

The disparity distribution fitting of SMG-SDF significantly improved the accuracy for US3D-Jacksonville, which had a disparity distribution notably different from that of the source domain. After WFG-SBF processing, the spectral brightness of the source domain was aligned with that of the target domain, further improving the accuracy of the HDADE in the target domain.

The 2-D and 3-D domain discriminators in FL-SSAdv had a significant positive impact on the generalization performance of HDADE, resulting in noticeable accuracy improvements for both target-domain test sets. IL-PCDS was more effective for the nonsame-source target-domain US3D-Jacksonville, where the domain shift was more pronounced.

Overall, the experimental results demonstrate that by gradually adding domain adaptation technical components, HDADE can progressively enhance the accuracy and generalization of the disparity estimation model. Ultimately, the fully integrated HDADE assisted the disparity estimation network in achieving the best disparity estimation results on both target-domain datasets.

## B. Sensitive Analysis of the Data Optimization

This work employs a data evaluation score threshold $\alpha$ for training data optimization. The selection of the $\alpha$-threshold critically governs training efficacy, necessitating systematic exploration of optimal $\alpha$ configurations. In this section, we benchmark the HMSM-Net using the WHU_Stereo-Qichun dataset as the source domain, training under 8 $\alpha$-thresholds ($\alpha \in [2.0, 9.0]$). Cross-domain validation is conducted on WHU_Stereo-5Cities and US3D-Jacksonville datasets to assess threshold rationality. Quantitative evaluations across thresholds are tabulated in Table V.

The experimental results reveal a critical tradeoff governed by the $\alpha$-threshold selection. Overly stringent thresholds ($\alpha < 4.0$) enhance the quality of data pairs but simultaneously diminish the training dataset

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

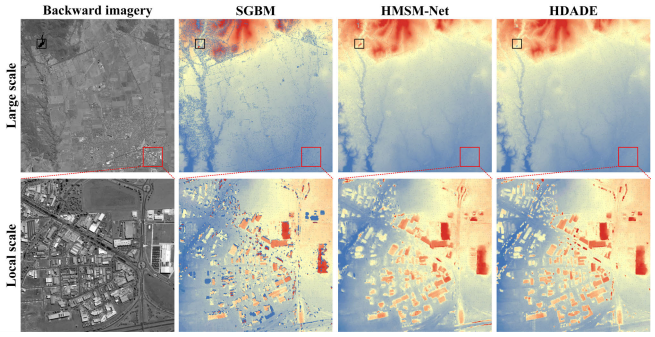



Fig. 16. Comparison of large-scale scenes across greater cross-domain shift.

volume to fewer than 451 samples. This volumetric insufficiency induces overfitting, manifesting as degraded cross-domain performance. Conversely, lenient thresholds ($\alpha > 8.0$) expand data diversity by incorporating over 553 training samples, effectively mitigating overfitting. However, this approach amplifies error propagation risks, increasing target-domain disparity estimation errors. Based on the quantitative validation in Table V, we select $\alpha = 5.0$ to optimally balance data quality and diversity, thereby ensuring robust generalization across domains.

### C. Large-Scale Transfer Experiments With a Greater Domain Shift

The image pairs in WHU_Stereo mainly consist of urban and rural scenes located in China. The applicability of disparity estimation methods, particularly deep learning-based methods, to different terrain types, architectural styles, and large-scale stereo satellite images has not yet been effectively explored. In this study, we used WHU_Stereo as the source-domain dataset and the constructed LargeS-GF7 as the target-domain dataset. Large-scale cross-domain experiments were conducted using SGBM, HMSM-Net, and HDADE methods. In the experiments, an area near Melbourne was selected as the test area. The images were rectified and cropped into stereo blocks, and a disparity estimation was performed for each block. Finally, the disparity blocks were stitched together to generate a DSM (results shown in Fig. 16).

From the overall perspective of the entire test area, SGBM exhibits a certain proportion of disparity estimation gaps, while the predictions from HMSM-Net and HDADE are smoother. In addition, both SGBM and the end-to-end HMSM-Net produced large areas of anomalously high values over the water bodies, as seen in the black border. HDADE significantly suppressed local anomalies, making the DSM more consistent with the stereo image content.

From a local perspective, as shown in the red box, the traditional SGBM method performed moderately well, with noticeable fragmentation and gaps at the edges of buildings. The original end-to-end HMSM-Net tends to miss small objects such as trees and low buildings, shows over-smoothing at building edges and produces anomalous elevations on roads. HMSM-Net processed with HDADE significantly improved these issues, exhibiting better detail for small objects such as trees and small buildings, more complete building contours without excessive smoothing at the edges, and smooth ground height variations.

In summary, the HDADE method proposed in this article has significant potential and practical value in large-scale satellite image stereo matching.

## VII. Conclusion

In this article, we propose a domain adaptation framework for deep learning-based OSSIDE methods called HDADE. It is designed to effectively mitigate the performance degradation of disparity estimation networks on cross-domain data. Although satellite stereo image disparity estimation has garnered significant research attention, the applicability of end-to-end methods to cross-domain and large-scale satellite images has not been thoroughly explored. Numerous studies have shown that the transferability of end-to-end disparity estimation methods fails to meet the application needs of OSSIDE tasks. HDADE constructs a comprehensive domain adaptation pipeline for end-to-end satellite image disparity estimation based on four levels: DL preprocessing, image-level coordinated augmentation, FL robust adversarial training, and instance-level self-training. HDADE significantly alleviates the performance degradation of disparity estimation networks in cross-domain satellite stereo data. Furthermore, HDADE has high plug-and-play versatility, thus allowing for its seamless integration with various types of disparity estimation networks.

Domain transfer experiments between WHU_Stereo and US3D datasets demonstrated that HDADE greatly enhanced the performance of disparity estimation algorithms on cross-domain data. Compared with other domain adaptation methods, HDADE exhibits clear accuracy advantages. Cross-domain experiments based on various networks further validate the generalizability of HDADE across multiple network types. On the global large-scale LargeS-GF7 validation dataset, HDADE produced the best visual results, demonstrating its significant potential for large-scale satellite image stereo matching applications. In summary, extensive cross-domain experiments involving various types of disparity estimation networks and multiple datasets with cross-payload, cross-spatiotemporal, and cross-stereo mode fully validated the effectiveness and flexibility of HDADE.

In the future, our research will combine domain generalization techniques to explore methods of achieving high generalization performance of disparity estimation networks trained in a single domain and applied across diverse domains. This will enable the more efficient and accurate adaptation of end-to-end disparity estimation techniques to diverse satellite image data, thereby providing more accurate disparity estimation results for practical application scenarios.

## Acknowledgment

The numerical calculations in this article have been done on the supercomputing system in the Supercomputing Center, Wuhan University, Wuhan, China.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

## REFERENCES


[1] M. Kukkonen, M. Maltamo, and P. Packalen, "Image matching as a data source for forest inventory-comparison of semi-global matching and next-generation automatic terrain extraction algorithms in a typical managed boreal forest environment," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 60, pp. 11–21, Aug. 2017, doi: 10.1016/j.jag.2017.03.012.

[2] M. Poggi, F. Tosi, K. Batsos, P. Mordohai, and S. Mattoccia, "On the synergies between machine learning and binocular stereo for depth estimation from images: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5314–5334, Sep. 2022, doi: 10.1109/TPAMI.2021.3070917.

[3] D.-G. Sim and R.-H. Park, "Localization based on DEM matching using multiple aerial image pairs," *IEEE Trans. Image Process.*, vol. 11, no. 1, pp. 52–55, Jan. 2002, doi: 10.1109/83.977882.

[4] M. Cord and D. Declercq, "Three-dimensional building detection and modeling using a statistical approach," *IEEE Trans. Image Process.*, vol. 10, no. 5, pp. 715–723, May 2001, doi: 10.1109/83.918565.

[5] H. Laga, L. V. Jospin, F. Boussaid, and M. Bennamoun, "A survey on deep learning techniques for stereo-based depth estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 1738–1764, Apr. 2022, doi: 10.1109/TPAMI.2020.3032602.

[6] H. Hirschmuller, "Stereo processing by semiglobal matching and mutual information," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 30, no. 2, pp. 328–341, Feb. 2008, doi: 10.1109/TPAMI.2007.1166.

[7] C. Barnes, E. Shechtman, A. Finkelstein, and D. B. Goldman, "PatchMatch: A randomized correspondence algorithm for structural image editing," *ACM Trans. Graph.*, vol. 28, no. 3, pp. 1–11, Jul. 2009, doi: 10.1145/1531326.1531330.

[8] G. Facciolo, C. D. Franchis, and E. Meinhardt, "MGM: A significantly more global matching for stereovision," in *Proc. Procedings Brit. Mach. Vis. Conf.*, Swansea, U.K., 2015, pp. 1–12.

[9] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Providence, RI, USA, Jun. 2012, pp. 3354–3361.

[10] N. Mayer et al., "A large dataset to train convolutional networks for disparity, optical flow, and scene flow estimation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 4040–4048.

[11] D. Scharstein and R. Szeliski, "High-accuracy stereo depth maps using structured light," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jul. 2003.

[12] J.-R. Chang and Y.-S. Chen, "Pyramid stereo matching network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 5410–5418.

[13] X. Song, X. Zhao, L. Fang, H. Hu, and Y. Yu, "EdgeStereo: An effective multi-task learning network for stereo matching and edge detection," *Int. J. Comput. Vis.*, vol. 128, no. 4, pp. 910–930, Apr. 2020, doi: 10.1007/s11263-019-01287-w.

[14] F. Zhang, V. Prisacariu, R. Yang, and P. H. S. Torr, "GA-net: Guided aggregation net for end-to-end stereo matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 185–194.

[15] J. Lou, W. Liu, Z. Chen, F. Liu, and J. Cheng, "ELFNet: Evidential local–global fusion for stereo matching," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Paris, France, Oct. 2023, pp. 17738–17747.

[16] A. Seki and M. Pollefeys, "SGM-nets: Semi-global matching with neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 6640–6649.

[17] C. Zhang, G. Meng, K. Tian, B. Ni, and S. Xiang, "Active disparity sampling for stereo matching with adjoint network," *IEEE Trans. Image Process.*, vol. 33, pp. 354–365, 2024, doi: 10.1109/TIP.2023.3343105.

[18] Z. Li et al., "Revisiting stereo depth estimation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 6177–6186.

[19] L. Zhao, Y. Liu, C. Men, and Y. Men, "Double propagation stereo matching for urban 3D reconstruction from satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601717, doi: 10.1109/TGRS.2021.3058144.

[20] S. He, S. Li, S. Jiang, and W. Jiang, "HMSM-net: Hierarchical multi-scale matching network for disparity estimation of high-resolution satellite stereo images," *ISPRS J. Photogramm. Remote Sens.*, vol. 188, pp. 314–330, Jun. 2022, doi: 10.1016/j.isprsjprs.2022.04.020.

[21] P. Liao et al., "$S^2$Net: A multitask learning network for semantic stereo of satellite image pairs," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5601313, doi: 10.1109/TGRS.2023.3335997.

[22] T. Wu, B. Vallet, M. Pierrot-Deseilligny, and E. Rupnik, "A new stereo dense matching benchmark dataset for deep learning," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2021, pp. 405–412, Jun. 2021, doi: 10.5194/isprs-archives-XLIII-B2-2021-405-2021.

[23] Y. Li, T. Shi, Y. Zhang, W. Chen, Z. Wang, and H. Li, "Learning deep semantic segmentation network under multiple weakly-supervised constraints for cross-domain remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 20–33, May 2021, doi: 10.1016/j.isprsjprs.2021.02.009.

[24] M. Luo and S. Ji, "Cross-spatiotemporal land-cover classification from VHR remote sensing images with deep learning based domain adaptation," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 105–128, Sep. 2022, doi: 10.1016/j.isprsjprs.2022.07.011.

[25] T. Wu, B. Vallet, M. Pierrot-Deseilligny, and E. Rupnik, "An evaluation of deep learning based stereo dense matching dataset shift from aerial images and a large scale stereo dataset," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 128, Apr. 2024, Art. no. 103715, doi: 10.1016/j.jag.2024.103715.

[26] S. Li, S. He, S. Jiang, W. Jiang, and L. Zhang, "WHU-stereo: A challenging benchmark for stereo matching of high-resolution satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603914, doi: 10.1109/TGRS.2023.3245205.

[27] J. Pang, W. Sun, J. Ren, C. G. Yang, and Q. Yan, "Cascade residual learning: A two-stage convolutional neural network for stereo matching," in *Proc. IEEE Int. Conf. Comput. Vis.*, Venice, Italy, Jan. 2017, pp. 878–886.

[28] Z. Liang et al., "Learning for disparity estimation through feature constancy," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 2811–2820.

[29] Z. Yin, T. Darrell, and F. Yu, "Hierarchical discrete distribution decomposition for match density estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 6037–6046.

[30] G. Yang, H. Zhao, J. Shi, Z. Deng, and J. Jia, "SegStereo: Exploiting semantic information for disparity estimation," in *Proc. Comput. Vis. (ECCV)*, Jan. 2018, pp. 660–676.

[31] X. Song, X. Zhao, H. Hu, and L. Fang, "EdgeStereo: A context integrated residual pyramid network for stereo matching," in *Proc. Comput. Vis. (ACCV)*, Cham, Switzerland, Jan. 2019, pp. 20–35.

[32] W. Zhan, X. Ou, Y. Yang, and L. Chen, "DSNet: Joint learning for scene segmentation and disparity estimation," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, Montreal, QC, Canada, May 2019, pp. 2946–2952.

[33] A. Kendall et al., "End-to-end learning of geometry and context for deep stereo regression," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 66–75.

[34] X. Guo, K. Yang, W. Yang, X. Wang, and H. Li, "Group-wise correlation stereo network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 3268–3277.

[35] G. Zhang et al., "A general deep learning framework guided by sparse matching for disparity estimation in high-resolution satellite stereo imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4507917, doi: 10.1109/TGRS.2024.3419781.

[36] W. Guo et al., "Context-enhanced stereo transformer," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, Jan. 2022, pp. 263–279.

[37] S. He, R. Zhou, S. Li, S. Jiang, and W. Jiang, "Disparity estimation of high-resolution remote sensing images with dual-scale matching network," *Remote Sens.*, vol. 13, no. 24, p. 5050, Dec. 2021, doi: 10.3390/rs13245050.

[38] Z. Shen, X. Song, Y. Dai, D. Zhou, Z. Rao, and L. Zhang, "Digging into uncertainty-based pseudo-label for robust stereo matching," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14301–14320, Dec. 2023, doi: 10.1109/TPAMI.2023.3300976.

[39] L. P. Osco et al., "A review on deep learning in UAV remote sensing," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 102, Oct. 2021, Art. no. 102456, doi: 10.1016/j.jag.2021.102456.

[40] C. Zhang, G. Meng, R. Y. D. Xu, S. Xiang, and C. Pan, "Learning adversarial point-wise domain alignment for stereo matching," *Neurocomputing*, vol. 491, pp. 564–574, Jun. 2022, doi: 10.1016/j.neucom.2021.12.034.

[41] T. Chang, X. Yang, T. Zhang, and M. Wang, "Domain generalized stereo matching via hierarchical visual transformation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, Jun. 2023, pp. 9559–9568.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.

[42] Z. Rao et al., "Masked representation learning for domain generalized stereo matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Vancouver, BC, Canada, Jun. 2023, pp. 5435–5444.

[43] X. Song, G. Yang, X. Zhu, H. Zhou, Z. Wang, and J. Shi, "AdaStereo: A simple and efficient approach for adaptive stereo matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Nashville, TN, USA, Jun. 2021, pp. 10323–10332.

[44] M. Poggi, D. Pallotti, F. Tosi, and S. Mattoccia, "Guided stereo matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 979–988.

[45] Z. Xu, Y. Li, S. Zhu, and Y. Sun, "Expanding sparse LiDAR depth and guiding stereo matching for robust dense depth estimation," *IEEE Robot. Autom. Lett.*, vol. 8, no. 3, pp. 1479–1486, Mar. 2023, doi: 10.1109/LRA.2023.3240093.

[46] Y.-K. Huang et al., "$S^3$: Learnable sparse signal superdensity for guided depth estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Nashville, TN, USA, Jun. 2021, pp. 16701–16711.

[47] B. Liu, H. Yu, and G. Qi, "GraftNet: Towards domain generalized stereo matching with a broad-spectrum and task-oriented feature," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, Jun. 2022, pp. 13002–13011.

[48] Y. Xia, P. d'Angelo, and J. Tian, "Dense matching comparison between classical and deep learning based algorithms for remote sensing data," *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. 43, no. B2, pp. 521–525, 2020, doi: 10.5194/ISPRS-ARCHIVES-XLIII-B2-2020-521-2020.

[49] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004, doi: 10.1023/B:VISI.0000029664.99615.94.

[50] H. Bay, A. Ess, T. Tuytelaars, and L. Gool, "Speeded-up robust features (SURF)," *Comput. Vis. Image Understand.*, vol. 110, no. 3, pp. 346–359, 2008, doi: 10.1016/j.cviu.2007.09.014.

[51] R. Liu, C. Yang, W. Sun, X. Wang, and H. Li, "StereoGAN: Bridging synthetic-to-real domain gap by joint optimization of domain translation and stereo matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 12754–12763.

[52] Q. Zhu, Z. Li, T. Song, L. Yao, Q. Guan, and L. Zhang, "Unrestricted region and scale: Deep self-supervised building mapping framework across different cities from five continents," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 344–367, Mar. 2024, doi: 10.1016/j.isprsjprs.2024.01.021.

[53] X. Li, M. Luo, S. Ji, L. Zhang, and M. Lu, "Evaluating generative adversarial networks based image-level domain transfer for multi-source remote sensing image segmentation and object detection," *Int. J. Remote Sens.*, vol. 41, no. 19, pp. 7343–7367, Oct. 2020, doi: 10.1080/01431161.2020.1757782.

[54] M. Poggi et al., "On the confidence of stereo matching in a deep-learning era: A quantitative evaluation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5293–5313, Sep. 2022, doi: 10.1109/TPAMI.2021.3069706.

[55] B. L. Saux et al. (Oct. 31, 2019). *Data Fusion Contest 2019 (DFC2019)*. [Online]. Available: https://ieee-dataport.org/open-access/data-fusion-contest-2019-dfc2019

[56] Y.-H. Tsai, W.-C. Hung, S. Schulter, K. Sohn, M.-H. Yang, and M. Chandraker, "Learning to adapt structured output space for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 7472–7481.

[57] J. Choi, T. Kim, and C. Kim, "Self-ensembling with GAN-based data augmentation for domain adaptation in semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6829–6839.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:35:33 UTC from IEEE Xplore. Restrictions apply.